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Software-Based Side-Channel Attacks in SDNs

Timing leakage in reactive OpenFlow networks

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SDN architecture and reactive forwarding

Layer	Role	Relevant state
Application	Defines network policy	Access policy and monitoring logic
Control	Computes and installs rules	Controller state and event workload
Data	Matches and forwards packets	Flow entries and rule lifetimes

Reactive forwarding can send an unmatched packet to the controller.

A table miss can also drop a packet or continue the pipeline, depending on configuration.

[1] ONF, OpenFlow v1.3.5 (2015)

The side channel is packet timing

Hidden network state can affect packet delay.

Hidden state	Timing observation
Rule present or absent	First-packet RTT difference
Rule expires	Delay changes over time
Controller does more work	Other control traffic slows

A software observer measures delay without directly reading the hidden state.

[2] Cui et al., IEEE TIFS (2016) [5] Yigit et al., Computer Networks (2023) [6] Sonchack et al., ACSAC (2016)

Attacker model and scope

Capabilities

Data-plane host or reachable endpoint

Software packet generation and timestamps

Repeated timing observations

Required conditions

State affects a measurable response

Traffic reaches the relevant switch or controller

Some techniques require header spoofing to be possible

Scope: OpenFlow timing leakage. CPU-cache attacks require a separate co-resident adversary model.

[2] Cui et al., IEEE TIFS (2016) [3] Liu, Reiter, and Sekar, ICDCS (2017) [4] Azzouni et al., GLOBECOM (2016)

Rule hits and controller-handled misses

Rule hit	Miss sent to the controller
An installed rule handles the packet	Packet-In requests a controller decision
No new controller round trip	Controller may install a rule and forward the packet

Miss delay ≈ Hit delay + Controller delay

Controller delay includes control-channel delay, queuing, and application processing. Network jitter can obscure this difference.

[1] ONF, OpenFlow v1.3.5 (2015) [2] Cui et al., IEEE TIFS (2016)

Flow-rule reconnaissance

01 **Observe candidate flow timings**

Use the packet path as a timing sensor.

02 **Infer likely rule hits or misses**

Estimate whether a covering rule already exists.

03 **Assess recent target activity**

Relate inferred rule state to a candidate flow.

Sources of ambiguity

Overlapping rules

Rule priorities

Expiry and eviction

Probe-induced changes

Rule-state inference requires context before attributing activity to a user.

[3] Liu, Reiter, and Sekar, ICDCS (2017)

Fingerprinting and rule timeouts

Inference target	Software observation	Interpretation limit
Network type	First versus later packet delay	Other network effects can mimic this gap
Controller family	Timing and response behavior	Needs known, comparable baselines
Rule timeouts	Delay changes after waiting	Noise and timer variation reduce certainty

Idle timeout measures inactivity.
Hard timeout bounds elapsed rule lifetime.

[1] ONF, OpenFlow v1.3.5 (2015) [4] Azzouni et al., GLOBECOM (2016) [5] Yigit et al., Computer Networks (2023)

Shared controller timing leakage

Configuration-dependent controller work can delay other control-plane traffic.

Sonchack et al. inferred host communication, ACL entries, and monitoring settings in a physical OpenFlow testbed.

>99%

Reported inference accuracy in the studied testbed

<50%

Reported accuracy with the timeout proxy

Published ACSAC 2016 results. These values describe the authors' experimental settings.

[6] Sonchack et al., ACSAC (2016)

Evaluation design for an SDN testbed

Proposed study

Component	Evaluation design
Testbed	Isolated OpenFlow switch with a reactive controller
Ground truth	Flow entries, expiry events, Packet-In counts, controller logs
Experimental factors	Controller load, jitter, rule overlap, software versus hardware switch
Metrics	ROC-AUC, false positives, probe budget, added p95 latency

Separate training and test runs. Compare with proactive forwarding and a non-SDN path. Validate emulation findings on hardware.

Defenses and their tradeoffs

Defense	Intended effect	Cost or residual risk
Proactive rules	Reduce reactive-installation gaps	More rule state and policy-update work
Delay normalization / proxy	Reduce state-dependent delays	Latency and fallback-policy constraints
Timeout randomization	Make expiry less predictable	Rule churn and repeated-observation risk
Isolation and probe limits	Reduce shared load and sampling	Resource costs and legitimate burst traffic

Engineering synthesis: evaluate leakage reduction together with latency and throughput.

[1] ONF, OpenFlow v1.3.5 (2015) [2] Cui et al., IEEE TIFS (2016) [5] Yigit et al., Computer Networks (2023) [6] Sonchack et al., ACSAC (2016)

Key takeaways

01 **Reactive control decisions can expose network state through timing.**

02 **Inference depends on rule semantics, topology, and background traffic.**

03 **Defense evaluation needs leakage metrics alongside network performance.**

Discussion

Which control decisions remain observable from the data plane?

[2] Cui et al., IEEE TIFS (2016) [3] Liu, Reiter, and Sekar, ICDCS (2017) [5] Yigit et al., Computer Networks (2023) [6] Sonchack et al., ACSAC (2016)

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[6] J. Sonchack et al. Timing-based Reconnaissance and Defense in Software-defined Networks.

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